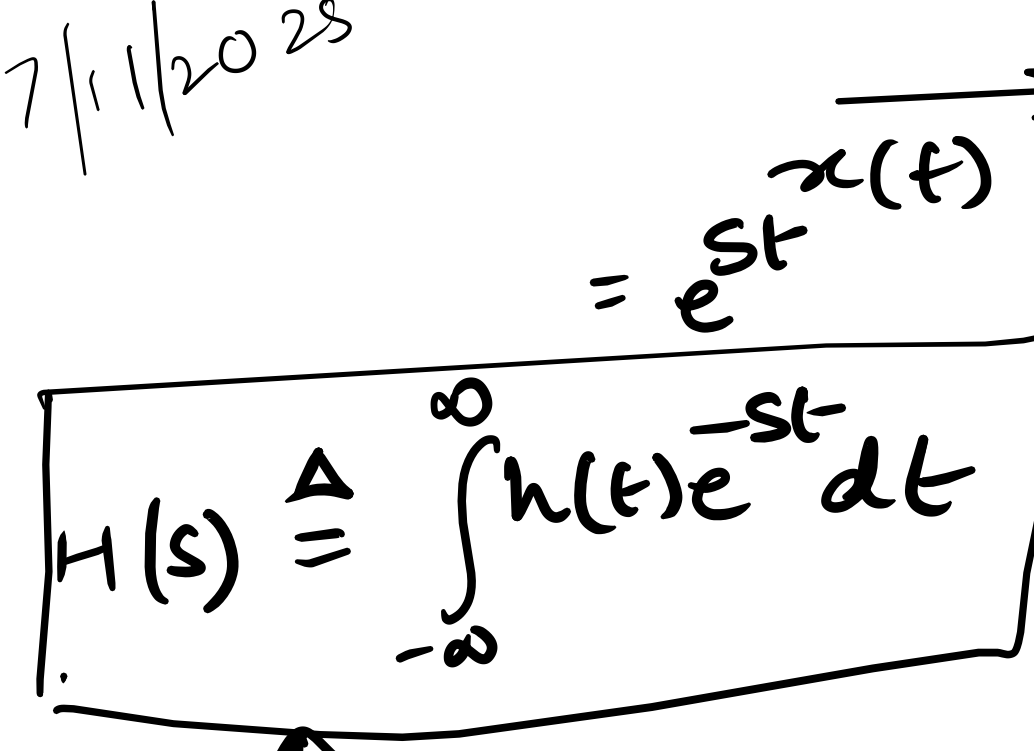
7/11/2025

Laplace Transform



$$= \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$$

$$= \int_{-\infty}^{\infty} e^{s(t-\tau)} h(\tau)\, d\tau$$

$$= e^{st} \int_{-\infty}^{\infty} h(\tau)\, e^{-s\tau}\, d\tau$$

$$= e^{st}\, H(s)$$

$$= e^{st}$$

$$H(s) \triangleq \int_{-\infty}^{\infty} h(t) e^{-st}\, dt$$

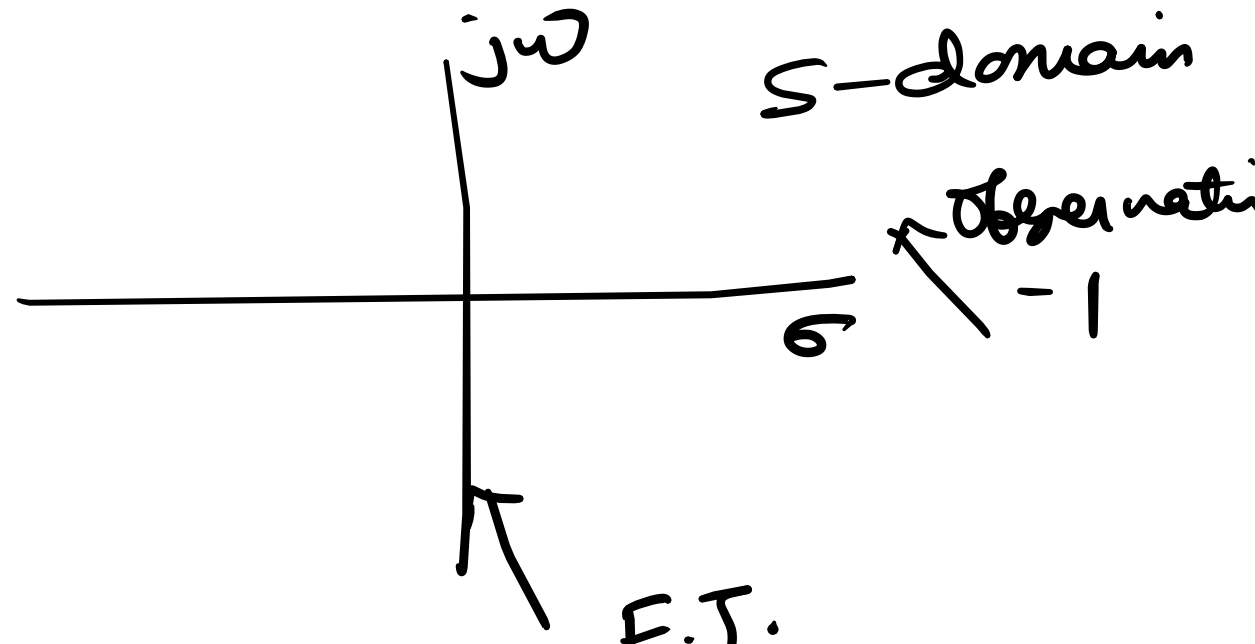
Definition of Laplace transform

① $s = j\omega$

$$H(j\omega) = \int_{-\infty}^{\infty} h(t)\, e^{-j\omega t}\, dt \Longleftarrow \text{F.T.}$$

Case-2 $s = \sigma + j\omega$ (σ: Re, $j\omega$: Im) $\Longleftarrow$ L.T.

Laplace is more generalized than F.T.



Laplace Domain

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st}\, dt$$

$$= \int_{-\infty}^{\infty} x(t) e^{-\sigma t} \cdot e^{-j\omega t}\, dt = \int_{-\infty}^{\infty} \left(x(t) e^{-\sigma t}\right) e^{-j\omega t}\, dt$$

$$= F.T.\left\{ x(t) e^{-\sigma t} \right\} \longleftarrow \text{Observation \#2}$$

$$x(t) = e^{-at}u(t)$$

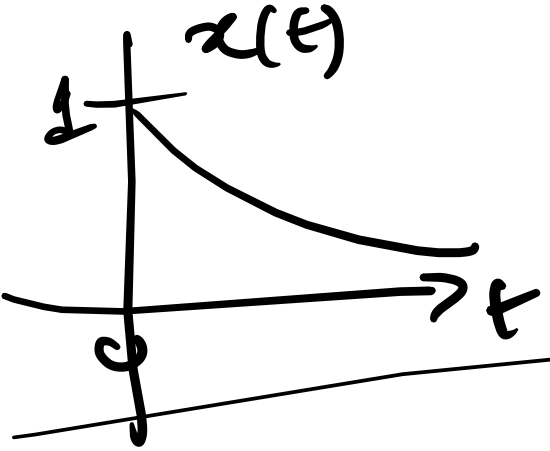
$$X(j\omega) = \int_{-\infty}^{\infty} \left(e^{-at}u(t)\right) e^{-j\omega t}\,dt$$

$$= \int_{0}^{\infty} e^{-at} e^{-j\omega t}\,dt = \int_{0}^{\infty} e^{-(a+j\omega)t}\,dt$$

$$= \frac{e^{-(a+j\omega)t}}{-(a+j\omega)}\Bigg|_{0}^{\infty} \longrightarrow \text{will converge to}$$

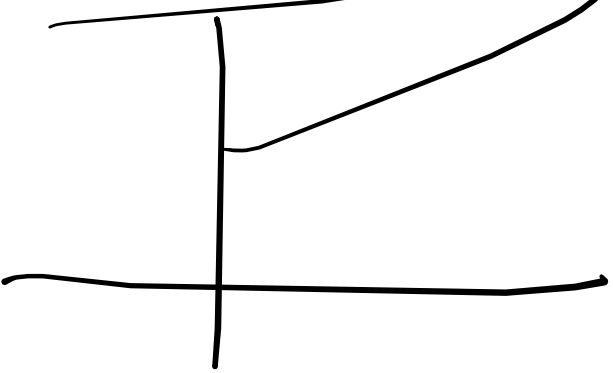
$$= \frac{1}{a+j\omega} \quad \text{if } a>0$$

for $a>0$

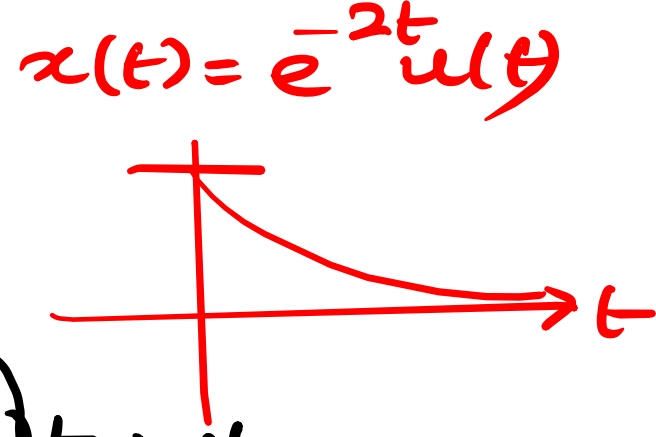


$\int_{-\infty}^{\infty} |x(t)|^2\,dt < \infty$

$a<0$



L.T. $x(t) = e^{-at}u(t)$



$$X(s) = \int_{-\infty}^{\infty} \left(e^{-at}u(t)\right) e^{-st}\,dt$$

$$= \int_{0}^{\infty} e^{-at} \cdot e^{-\sigma t} \cdot e^{-j\omega t}\,dt = \int_{0}^{\infty} e^{-(a+\sigma)t - j\omega t}\,dt$$

$$= \left. \frac{e^{-(a+\sigma)t - j\omega t}}{-(a+\sigma+j\omega)} \right|_{0}^{\infty} = \frac{1}{s+a}$$

if $a+\sigma > 0$

$\sigma > -a$

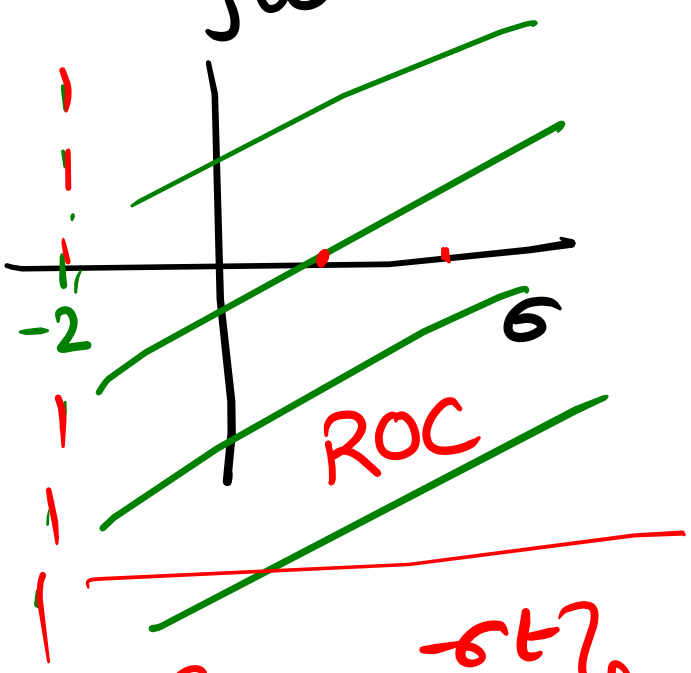
or

or $\boxed{\text{Re}(s) > -a}$

(ROC) $\Leftarrow$ Region of convergence

$a>0$

$a=2$



F.T. $\{x(t)e^{-\sigma t}\}$

F.T. $\{e^{-at}u(t)e^{-\sigma t}\}$

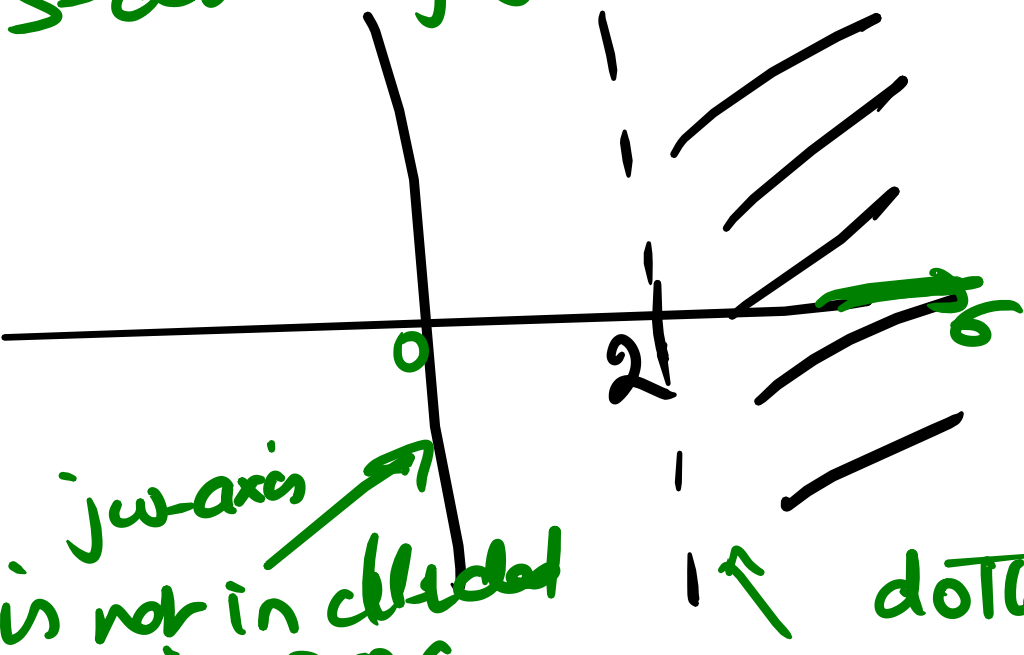
if ROC includes jω-axis
F.T. of signal exists

if ROC does not include jω-axis
F.T. of signal does not exist.

Let- $x(t) = e^{2t}u(t)$ → Compare with e^{-at}

$a = -2$

S-domain jω



$Re(s) > -a$

$\Rightarrow$ $Re(s) > 2$

Strictly inequality $\sigma = 2$

jω-axis is not included in ROC
$\Rightarrow$ F.T. of $x(t)$ does not exist

dotted line to exclude $\sigma = 2$

$$x(t) = e^{-at}u(t)$$

$$X(s) = \frac{1}{a+s}\,; \quad Re(s) > -a$$

Laplace

Let $a = 0$

$$X(s) = \frac{1}{s}\,; \quad Re(s) > 0$$

$$x(t) = u(t)$$

